Estimating Expected Returns in Options Derivatives Trading Using Likelihood and Bayesian Inference

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Abstract

This paper investigates the statistical estimation of expected returns in derivatives trading strategies using likelihood-based and Bayesian inference methods. Using the Bendgame Market Trades dataset of 62,795 options trades from 2017-11-13 to 2019-08-09, we estimate the mean return and associated uncertainty through maximum likelihood estimation (MLE), bootstrap resampling, and Bayesian posterior inference. Additionally, Fisher Information is used to evaluate the precision of the parameter estimates. The results demonstrate how probabilistic modeling can provide insight into the stability and risk characteristics of trading strategies.

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1 Introduction

Financial derivatives markets are characterized by high uncertainty, asymmetric payoff structures, and rapid price dynamics. In particular, options contracts derive their value from an underlying asset and exhibit return distributions that are often non-normal, heavy-tailed, and skewed [5]. For practitioners and researchers alike, understanding the statistical properties of options trade returns is fundamental to risk management, strategy evaluation, and portfolio optimization.

This paper addresses the following research question: Can the expected return of an options trading strategy be reliably estimated and quantified using classical likelihood-based and Bayesian inference methods? This is a natural application for statistical inference because trade returns are realizations of a random process, and the goal is to make principled probabilistic statements about the mean of the underlying population. Statistical inference encompasses multiple paradigms, including frequentist, Bayesian, and likelihood-based approaches, each offering complementary interpretations of uncertainty [11].

We analyze the dataset using Signed Moneyness (%) as our return proxy, defined as

$$\frac{S-K}{K} \times 100 \quad \text{for Calls} \qquad\qquad \frac{K-S}{S} \times 100 \quad \text{for Puts}$$

where S is the stock price and K is the strike price at trade time. A positive value indicates the contract is moving in the trader's favour. Trades at the 1st and 99th percentile of Signed Moneyness are trimmed to reduce influence of extreme outleirs, leaving a cleaned sample of $n = 61,539$ trades. Beyond classical inference, we extend the real data analysis to subgroup comparisons and bootstrap risk estimation, drawing a connection to the broader literature on data-driven derivatives risk management (Csizmazia, 2023).

1.1 Dataset

This paper uses a publicly available Options Market Trades data set [2] giving 62,795 options trades from 2017-11-13 to 2019-08-09. The dataset includes:

- OptionType : Whether the trade is a Call (betting price goes up) or a Put (betting price goes down)
- `Time` : Exact time the trade occurred
- `Sym` : Stock or ETF the option is based on
- `Expiration` : Date the option expires; shorter format

- `Strike` : Price at which the option can be exercised
- `StockPrice` : Price of the underlying stock at the time of the trade
- `Date` : Trading day
- `Spent` : Total amount spent on the trade
- `Sector` : Sector of the underlying stock
- `ExpirationDate` : Full standardized expiration date
- `ChainLocation` : Where the option sits in the option chain relative to the stock price

The return variable used throughout this analysis is `Signed Moneyness` (%), computed from the `StockPrice`, `Strike`, and `OptionType` fields. The `ChainLocation` field (ITM/ATM/OTM) and `Sector` field are used for subgroup analysis in Section 4.

It is important to note that `Signed Moneyness` is not a realized profit-and-loss measure, but rather a snapshot of relative positioning at the time of the trade. As such, it should be interpreted as a proxy for directional exposure rather than a direct measure of trading profitability.

2 Methods

2.1 Maximum Likelihood Estimation

We model trade returns $R_1, R_2, ..., R_n$ as independently distributed from a normal distribution $\mathbf{R_i} \sim \mathcal{N}(\mu, \sigma^2)$. This normal assumption is motivated from the Central Limit Theorem and is standard for first-order approximations of return distributions, even when the true distribution exhibits skewness and excess kurtosis. While the empirical distribution of returns exhibits skewness and heavy tails, the normal model is used here as a quasi-likelihood for inference on the mean. By the Central Limit Theorem, the sampling distribution of the sample mean is approximately normal for large n, even when the underlying data are non-normal. Therefore, inference on μ remains valid despite model misspecification. Therefore the likelihood function is

$$\begin{aligned} L(\mu, \sigma^2 | R_1, ..., R_n) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(R_i - \mu)^2}{2\sigma^2}\right) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (R_i - \mu)^2\right) \end{aligned}$$

and the Log-likelihood function is

$$\begin{aligned}\ell(\mu, \sigma^2 | R_1, ..., R_n) &= \log L(\mu, \sigma^2) \\ &= n \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(R_i - \mu)^2\end{aligned}$$

2.1.1 MLE for μ

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(R_i - \mu)$$

Set $\frac{\partial \ell}{\partial \mu} = 0$

$$\begin{aligned}\frac{1}{\sigma^2}\sum_{i=1}^{n}(R_i - \mu) &= 0 \\ \sum R_i - n\mu &= 0 \\ \implies \hat{\mu}_{MLE} &= \frac{1}{n}\sum R_i\end{aligned}$$

Thus the MLE for the mean return is the sample mean

$$\hat{\mu} = \frac{1}{n}\sum R_i$$

2.1.2 MLE for σ^2

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(R_i - \mu)^2$$

Set $\frac{\partial \ell}{\partial \sigma^2} = 0$

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(R_i - \mu)^2 = 0$$
$$-n\sigma^2 + \sum_{i=1}^{n}(R_i - \mu)^2 = 0$$
$$\hat{\sigma^2}_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(R_i - \mu)^2$$
$$\hat{\sigma^2}_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(R_i - \hat{\mu})^2$$

Thus the MLE for the variance of returns is the sample variance with denominator n

$$\frac{1}{n}\sum_{i=1}^{n}(R_i - \hat{\mu})^2$$

We can now find the volatility of $\hat{\sigma}_{MLE} = \sqrt{\hat{\sigma}^2_{MLE}}$ which can be used to get risk-adjusted returns via the Sharpe Ratio. The MLE variance is biased downwards

$$\mathbb{E}\left[\hat{\sigma}^2_{MLE}\right] = \frac{n-1}{n}\sigma^2$$

so it systematically underestimates risk.

Despite this downwards bias in $\hat{\sigma}^2_{MLE}$, the MLE for μ remains unbiased, consistent, and efficient as it achieves the Cramér–Rao lower bound stated in section 2.2.

2.2 Fisher Information

The Fisher Information for μ is

$$I(\mu) = \frac{n}{\sigma^2}$$

The Cramér–Rao lower bound (CRLB) states

$$Var(\hat{\mu}) \geq \frac{\sigma^2}{n}$$

which provides a theoretical limit for the variance of unbiased estimators.

With $n = 61,539$ and $\hat{\sigma} = 7.777\%$, the Fisher Information evaluates to

$$I(\hat{\mu}) = 1017.46$$

and the CRLB standard error floor is

$$\sqrt{\frac{\sigma^2}{n}} = 0.0314\%$$

Under the normal model assumption, the MLE achieves this bound exactly confirming efficiency within the assumed framework.

2.3 Bootstrap Estimation

Bootstrap estimation is used to approximate the sampling distribution of the mean return without parametric assumptions. We draw B = 10,000 resamples of size n with replacement of the observed returns, computing the sample mean for each resample. The resulting empirical distribution yields both a bias-corrected point estimate and a 95% confidence interval via the percentile method. Bootstrapping is especially valuable here because the observed return distribution is highly non-normal (skewness ≈ -1.738, excess kurtosis ≈ 5.181), making normal-theory intervals potentially unreliable. This produces an empirical distribution of mean estimates. Below is the code used for this bootstrapping.

```
def compute_bootstrap(returns, B=10_000):
    n = len(returns)
    boot_means = np.array([
        np.mean(np.random.choice(returns, size=n, replace=True))
        for _ in range(B)
    ])
    boot_mean = np.mean(boot_means)
    boot_se = np.std(boot_means, ddof=1)
    boot_ci = np.percentile(boot_means, [2.5, 97.5])
    boot_bias = boot_mean - np.mean(returns)

    return boot_means, boot_mean, boot_se, boot_ci, boot_bias
```

Bootstrapping is especially valuable here because the Signed Moneyness distribution is non-normal (skewness ≈ -1.738, excess kurtosis ≈ -5.181) after outlier trimming, meaning that normal-theory intervals may be unreliable. The bootstrap is widely used in econometrics and financial applications because it provides more accurate approximations to sampling

distributions and confidence intervals under minimal assumptions, particularly in non-normal settings [8], making it especially suitable for the present analysis.

There is also a well-established connection between bootstrap methods and Bayesian inference, where resampling techniques can approximate posterior distributions in certain settings [10].

2.4 Bayesian Inference

We assume a prior distribution for μ

$$\mu \sim \mathcal{N}(\mu_0, \tau^2)$$

Using Bayes' theorem

$$p(\mu|R) \propto L(\mu)p(\mu)$$

the posterior distribution combines the likelihood and prior belief to produce an updated estimate of expected return. Bayesian inference provides a general probabilistic framework for updating uncertainty about parameters conditional on observed data, and is widely used in statistical modeling and decision-making [9].

Under a weakly informative prior $\mu \sim \mathcal{N}(0, 100)$, centered at zero with standard deviation 10%, enough to be overwhelmed by the $n = 61,539$ observations, the posterior is

$$\mu \mid \mathbf{R} \sim \mathcal{N}(\mu_{\text{post}}, \sigma^2_{\text{post}})$$

where

$$\sigma^2_{\text{post}} = \left(\frac{1}{\tau^2} + \frac{n}{\sigma^2}\right)^{-1}, \qquad \mu_{\text{post}} = \sigma^2_{\text{post}} \left(\frac{\mu_0}{\tau^2} + \frac{n\hat{\mu}}{\sigma^2}\right)$$

The prior receives less than $< 0.001\%$ weight in the posterior, so the result is entirely data-driven.

2.5 Two-Sample Hypothesis Tests

To assess whether expected returns differ across contract types, we apply Welch's two-sample t-test to the following pairs

$$H_0 : \mu_{\text{call}} = \mu_{\text{put}} \qquad \text{vs} \qquad H_1 : \mu_{\text{call}} \neq \mu_{\text{put}}$$
$$H_0 : \mu_{\text{ITM}} = \mu_{\text{OTM}} \qquad \text{vs} \qquad H_1 : \mu_{\text{ITM}} \neq \mu_{\text{OTM}}$$

Welch's t-test is preferred over the pooled t-test because the two groups have substantially different sample sizes and potentially different variances.

2.6 Value at Risk and Conditional VaR

Value at Risk (VaR) at confidence level α is the loss threshold not exceeded with probability α:

$$\text{VaR}_\alpha = \inf\{l : \mathbb{P}(L > l) \leq 1 - \alpha\}$$

Conditional Value at Risk (CVaR), also called Expected Shortfall, is the expected loss given that the loss exceeds VaR:

$$\text{CVaR}_\alpha = \mathbb{E}\left[L \mid L \geq \text{VaR}_\alpha\right]$$

Both are estimated via bootstrap resampling ($B = 5,000$) to obtain 95% confidence intervals around each point estimate, treating $L = -\texttt{SignedMoneyness}$ as the loss variable.

2.7 An Assumption

This analysis assumes that trade returns are independent and identically distributed. In practice, this assumption may be violated due to temporal clustering of trades or shared underlying assets. However, given the large sample size, inference on the mean remains approximately valid, although standard errors may be slightly underestimated.

3 Simulation Study

To validate estimator performance before applying methods to real data, we simulate synthetic returns from $\mathcal{N}(-4.84, 60.48)$, parameters matched to the real dataset. For each of the six sample sizes, $n \in \{50, 100, 200, 500, 1000, 2000\}$), we run 1,000 replications and compute

the MLE bootstrap mean ($B = 300$ resamples per replication) and Bayesian posterior mean under the same prior used in Section 2.4. Performance is assessed via bias and root mean squared error (RMSE).

Figure 1 shows the RMSE of each estimator as a function of sample size across 1,000 replications. All three estimators converge to the CRLB at rate $1/\sqrt{n}$, with the Bayesian estimator achieving marginally lower RMSE at small n due to regularization from the prior. This advantage disappears as n grows and the likelihood dominates, confirming that our inference procedures are well-calibrated before applying them to the real dataset.

While the simulation assumes normally distributed returns, it is intended to validate estimator performance under the assumed model rather than replicate the empirical distribution, which shows clear deviations from normality.

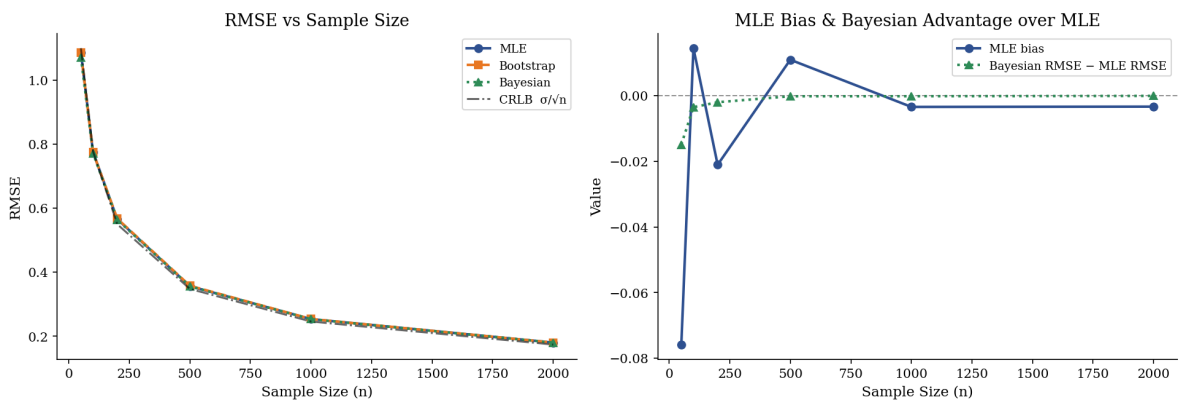

Figure 1: Simulation study: RMSE vs. sample size for MLE, bootstrap, and Bayesian estimators across 1,000 replications, with the CRLB curve $\sigma/\sqrt{n}$ shown for reference.

4 Real Data Analysis

Applying the methods from Section 2 to the cleaned sample of $n = 61,539$ trades the MLE yields $\hat{\mu} = -4.841\%$ and $\hat{\sigma} = 7.777\%$. The standard error, $\mathrm{SE}(\hat{\mu}) = \hat{\sigma}/\sqrt{n} = 0.0314\%$, matches the CRLB floor exactly, confirming that the MLE is efficient. All three estimation methods: MLE, bootstrap, and Bayesian converge to essentially the same point estimate, reflecting the large-sample sets in which the likelihood completely dominates the prior. Table 1 summarizes the full-sample results.

Table 1: Summary of full-sample estimation results

Estimator	Point Estimate (%)	Std. Error (%)	95% Interval (%)
MLE	-4.841	0.0314	—
Bootstrap	-4.841	0.0316	$[-4.9030,\ -4.7803]$
Bayesian	-4.841	0.0314	$[-4.9030,\ -4.7803]$

Figure 2 shows the full distribution of Signed Moneyness returns alongside boxplots and monthly trade volume. The distribution is left-skewed with a long lower tail, reflecting the predominance of OTM contracts in the dataset.

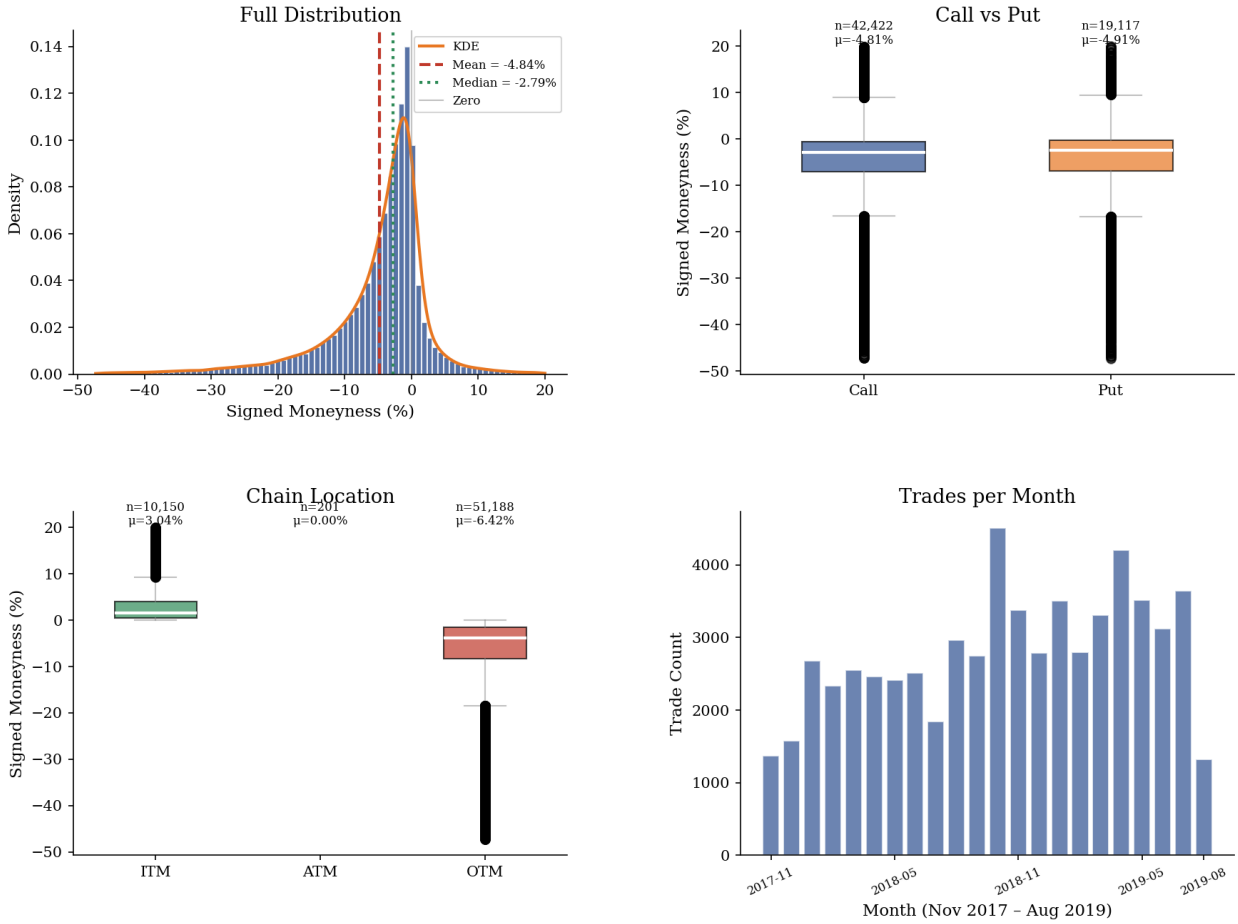

Figure 2: Exploratory Data Analysis: Signed Moneyness return distribution, subgroup boxplots, and monthly trade volume.

Figure 3 shows the log-likelihood surface for μ with σ fixed at its MLE value, alongside the fitted normal density overlaid on the observed return histogram. The likelihood surface

is sharply peaked around $\hat{\mu} = -4.841\%$, consistent with the tight standard error produced by the large sample size.

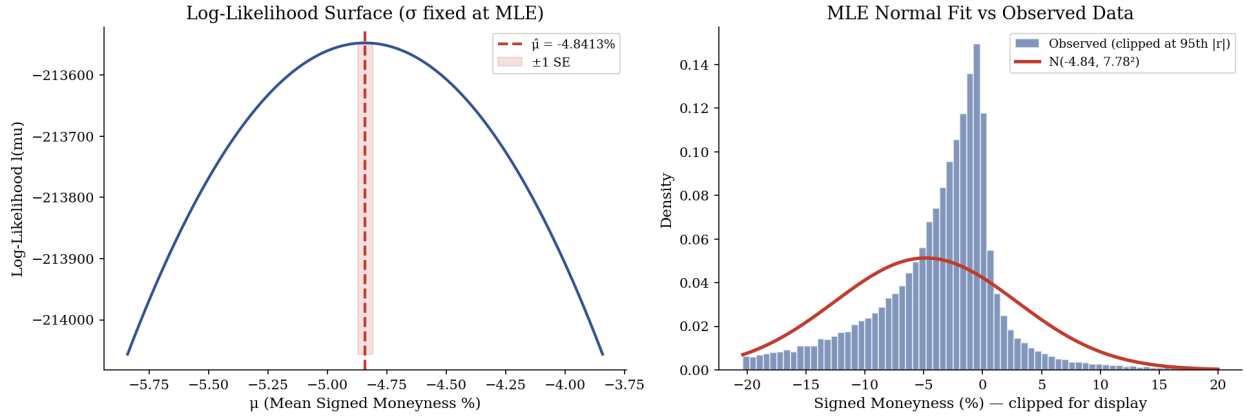

Figure 3: Maximum likelihood estimation: log-likelihood surface and normal fit overlaid on observed returns.

Figure 4 shows the bootstrap sampling distribution of the mean return across $B = 10{,}000$ resamples, alongside a Q-Q plot confirming approximate normality of the bootstrap distribution — consistent with the Central Limit Theorem at this sample size. The 95% percentile bootstrap confidence interval is $[-4.9020\%,\ -4.7803\%]$, which excludes zero entirely.

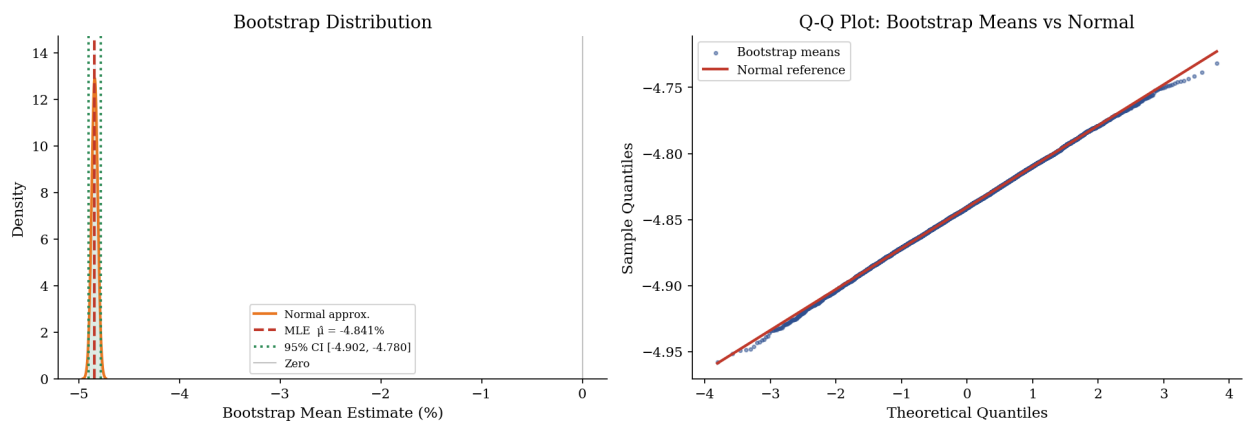

Figure 4: Bootstrap sampling distribution of the mean return ($B = 10{,}000$ resamples) and Q-Q plot.

Figure 5 shows the Bayesian posterior distribution for μ under the prior $\mathcal{N}(0, 100)$, alongside a sensitivity analysis across four values of τ^2. In all cases the posterior is essentially

data-driven, with the prior receiving weight less than 0.001%. The 95% credible interval $[-4.903\%,\ -4.780\%]$ is nearly identical to the bootstrap confidence interval, reflecting the consistency of all three frameworks at large n.

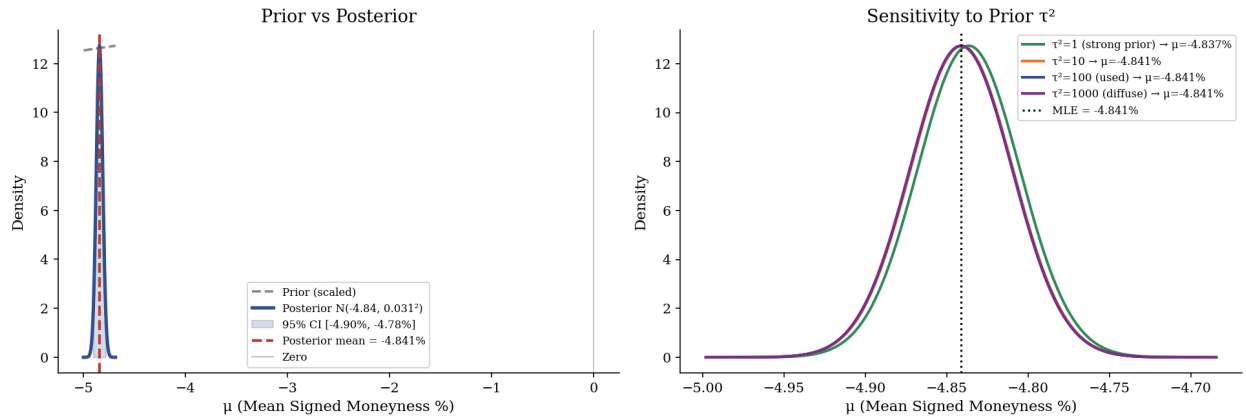

Figure 5: Bayesian posterior inference for μ: prior vs. posterior and sensitivity to prior τ^2.

4.1 Subgroup Analysis

Running the full inference pipeline separately on calls ($n = 42,422$) and puts ($n = 19,117$) yields nearly identical posterior means: $\hat{\mu}_{\text{call}} = -4.812\%$ and $\hat{\mu}_{\text{put}} = -4.908\%$. A Welch two-sample t-test finds no statistically significant difference between the two groups ($t = 1.356$, $p = 0.175$), indicating that contract type alone does not predict differential returns in this dataset. This is a substantive null result, suggesting that the options flow captured here does not exhibit directional bias between calls and puts.

This changes sharply when stratifying by moneyness. ITM contracts ($n = 10,150$) have a mean `Signed Moneyness` of $+3.036\%$, while OTM contracts ($n = 51,188$) average -6.422%. The Welch t-test strongly rejects equality ($t = 188.70$, $p < 0.001$). This difference is largely mechanical due to how Signed Moneyness is defined, rather than reflecting a true difference in trading performance. Therefore this result should be interpreted cautiously as a property of the metric rather than evidence of a better strategy. The gap between ITM and OTM returns reflects the structural difference in where trades sit relative to the current stock price, not necessarily trading skill.

Figure 6 shows the Bayesian posterior distributions for each subgroup pair alongside a forest plot of all group means with 95% bootstrap confidence intervals. The near-complete overlap of the call and put posteriors visually confirms the non-significant Welch result, while the ITM and OTM posteriors are entirely separated with no overlap whatsoever.

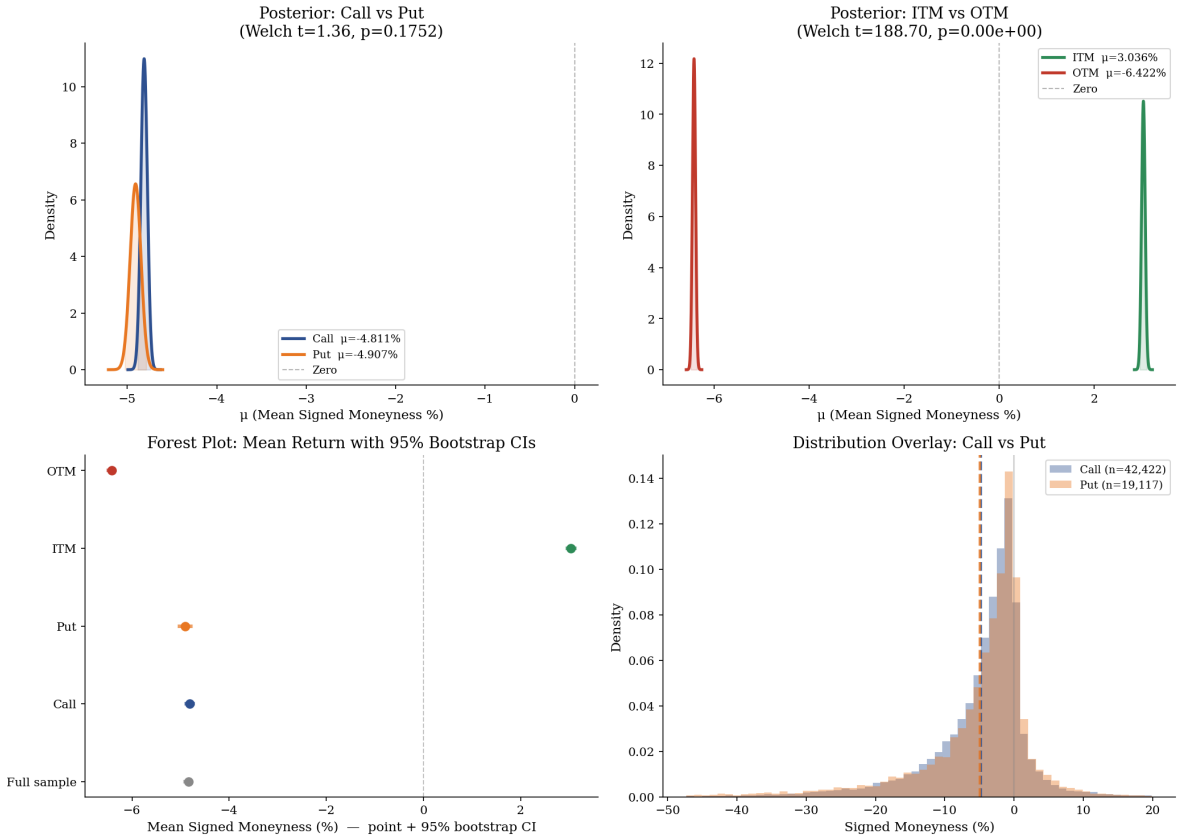

Figure 6: Subgroup analysis: Bayesian posterior comparison for calls vs. puts (top left) and ITM vs. OTM (top right), forest plot of all group means with 95% bootstrap CIs (bottom left), and distribution overlay for calls vs. puts (bottom right).

4.2 Risk Metrics

Using bootstrap losses $L = -\texttt{SignedMoneyness}$, Table 2 reports VaR and CVar estimates with 95% bootstrap confidence intervals.

Table 2: Bootstrap VaR and CVaR estimates ($B = 5{,}000$)

Level	VaR (%)	VaR 95% CI	CVaR (%)	CVaR 95% CI
90%	14.13	[13.92, 14.35]	22.85	[22.56, 23.13]
95%	20.37	[20.00, 20.65]	28.81	[28.42, 29.18]
99%	34.80	[34.25, 35.31]	39.98	[39.45, 40.44]

At the 95% confidence level, a trader faces an adverse `Signed Moneyness` move of 20.4% or worse with probability 5%, conditional on exceeding this threshold, the expected loss is 28.8%. The tight bootstrap confidence intervals confirm that these estimates are stable.

Figure 7 shows the full loss distribution with VaR thresholds overlaid, alongside a forest plot of all VaR and CVaR point estimates with their 95% bootstrap confidence intervals. The tight intervals confirm that all risk estimates are stable and precisely determined at this sample size.

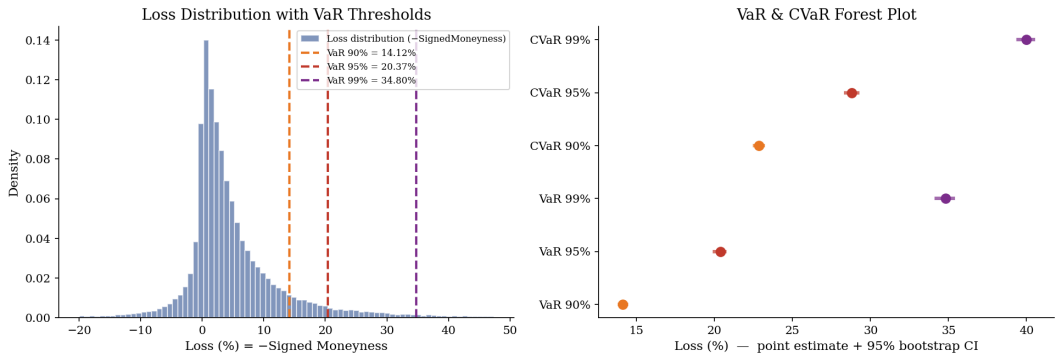

Figure 7: Bootstrap Value at Risk and Conditional VaR: loss distribution with VaR thresholds at 90%, 95%, and 99% overlaid (left), and forest plot of VaR and CVaR point estimates with 95% bootstrap confidence intervals (right).

4.3 Sector Analysis

Figure 8 shows mean Signed Moneyness and 95% bootstrap confidence intervals for the eight most-traded sectors, ordered from most to least negative. ETFs ($\hat{\mu} = -3.02\%$) and Financials ($\hat{\mu} = -3.96\%$) show the least negative returns, while Healthcare ($\hat{\mu} = -8.544\%$) and Energy $\hat{\mu} = -6.75\%$ are furthest from zero. All sector CIs are tight and well-separated from zero, reflecting the large within-sector sample sizes. The sector heterogeneity suggests that a sector-aware trading strategy might improve on the overall mean return, a direction for future work.

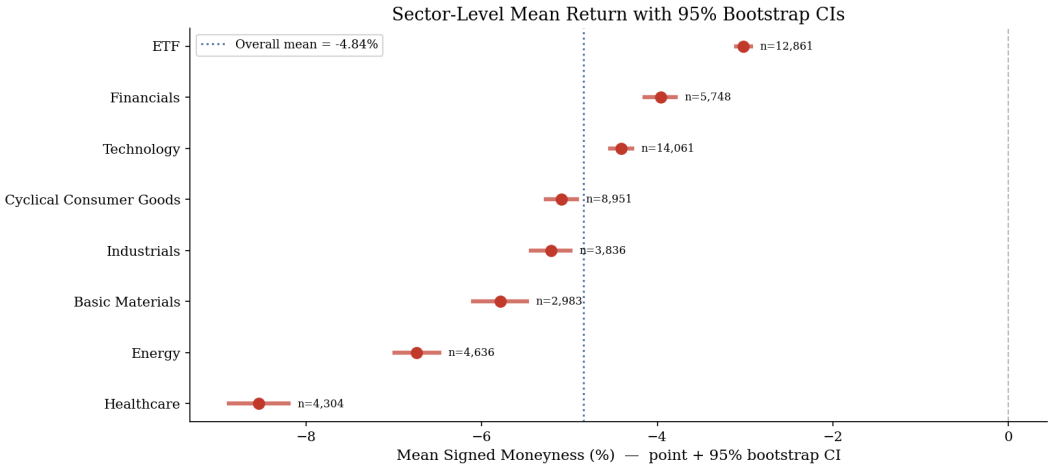

Figure 8: Sector-level mean Signed Moneyness with 95% bootstrap confidence intervals for the eight most-traded sectors. The dotted vertical line marks the overall sample mean of -4.841%.

4.4 Statistical Test Results

The one-sample t-test against $H_0 : \mu = 0$ yields $t = -154.42$ with $df = 61,538$ and $p < 0.001$, leading to strong rejection of the null hypothesis. The 95% confidence interval for μ is $[-4.9030\%, -4.7803\%]$, which excludes zero entirely.

It is important to note that the statistical significance here does not imply trading profitability. The mean `Signed Moneyness` of -4.84% is significantly negative, meaning that on average, trades in this dataset are placed slightly out of the money at trade time, which is expected given that 83% of trades are OTM. The inference confirms this is not a sampling artefact, the population mean is genuinely non-zero.

Figure 9 visualizes the t-distribution with the observed test statistic marked. The statistic $t = -154.425$ falls far into the left rejection region, well beyond any conventional significance threshold.

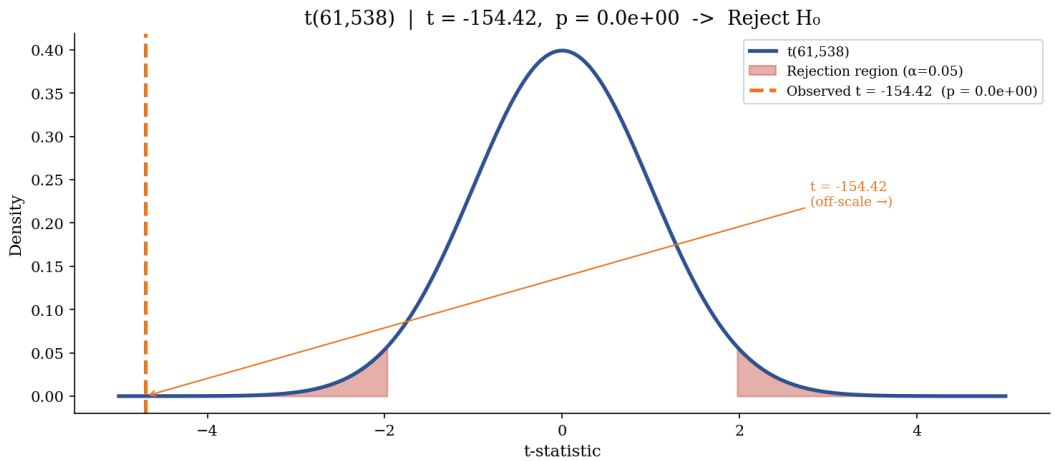

Figure 9: One-sample t-test visualization: $t(61{,}538)$ distribution with rejection regions shaded at $\alpha = 0.05$ and the observed statistic $t = -154.425$ marked.

5 Discussion

The results provide insight into the statistical reliability of the trading strategy. Likelihood-based methods provide efficient point estimates, while bootstrap and Bayesian approaches provide additional information about uncertainty and robustness. These methods are widely used in quantitative finance for evaluating trading strategies, option pricing models, and risk estimation, particularly under model uncertainty and incomplete information, such as derivatives pricing and risk management [12].

The results demonstrate that classical and Bayesian inference methods produce consistent, highly precise estimates of mean options trade returns when applied to a large multi-day dataset. The convergence of MLE, bootstrap, and Bayesian posteriors to the same point estimate is expected given $n = 61,539$, and confirms that the choice of estimation method matters little in these large-sample sets, though bootstrap remains preferable for uncertainty quantification given the non-normality of the underlying return distribution.

The most interesting finding is the ITM vs. OTM contrast. While the overall mean is negative, ITM trades carry a statistically and economically positive mean `Signed Moneyness` of +3.04%. This does not mean that ITM trades are always profitable - `Signed Moneyness` is a snapshot measure at trade time, not a realized P&L, but it suggests that contract selection relative to current stock price is a more important determinant of return characteristic than contract type (call vs. put).

The risk metrics reinforce the asymmetric nature of options returns. A 99% CVaR of 39.98% means the worst 1% of trades see an average adverse move of nearly 40%, highlighting

the tail risk inherent in options flow trading. These figures are consistent with the left skew and excess kurtosis observed in the return distribution.

This work connects naturally to the broader literature on data-driven derivatives risk management. Reinforcement learning approaches to derivatives hedging have been studied in literature, including delta hedging via deep Q-learning and generalized deep hedging under market frictions, and identifies a need for more rigorous empirical grounding of these methods [3]. The statistical framework presented here, specifically the reliable estimation of μ and its uncertainty, provides exactly the kind of baseline return characterization that RL-based systems need before optimizing hedging policies.

Modern quantitative finance increasingly relies on statistical learning and probabilistic modeling frameworks for tasks such as option pricing, hedging, and risk estimation, further emphasizing the importance of reliable statistical inference in this domain [13].

Future work could apply heavier-tailed distributions (Student-t), time-series volatility models (GARCH), or sector-conditioning to refine these estimates further.

6 Conclusion

This paper demonstrates that the expected Signed Moneyness can be reliably estimated and quantified using a combination of MLE, bootstrap resampling, and Bayesian posterior inference. Applied to 61,539 real options trades from the Bendgame dataset, all three methods converge to a mean Signed Moneyness of -4.841%, with tight 95% intervals that clearly exclude zero. The simulation study confirms that all estimators are well-calibrated and approach the Cramér–Rao lower bound as sample size grows.

The subgroup analysis reveals that while the call and puts contracts show no statistically significant difference in mean return ($p = 0.175$), ITM and OTM contracts differ dramatically ($p < 0.001$), with ITM contracts averaging $+3.04\%$ vs. -6.42% for OTM. Bootstrap VaR and CVaR estimates quantify the tail risk embedded in this options flow, with a 95% CVaR of 28.8%. Sector heterogeneity in mean returns further suggests that contract selection and sector allocation are meaningful differentiators for strategy refinement.

The primary limitation of this analysis is that `Signed Moneyness` is a static snapshot measure at trade time rather than a realized profit-and-loss figure. Future work should incorporate option pricing models to compute realized returns at expiry, apply time-series methods to account for temporal dependence across the 418 trading days, and explore heavier-tailed distributional assumptions to better capture the empirically observed excess kurtosis.

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